
Mechanistic Interpretability of EEG Foundation Models via Sparse Autoencoders

William Lehn-Schiøler^{1,2}, Magnus Ruud Kjær², Rahul Thapa³, Magnus Guldberg Pedersen^{1,4}, Anton Storgaard Mosquera^{1,4}, Nick Williams⁵, Andreas Brink-Kjær², Tue Lehn-Schiøler¹, Sándor Beniczky^{6,7}, Lars Kai Hansen⁴, Sadasivan Puthusserypady²

¹BrainCapture, Copenhagen, Denmark

²DTU Health Tech, Technical University of Denmark, Kongens Lyngby, Denmark

³Stanford University, Stanford, CA, USA

⁴DTU Compute, Technical University of Denmark, Kongens Lyngby, Denmark

⁵Seer Medical, Melbourne, Australia

⁶Filadelfia Epilepsy Hospital, Dianalund, Denmark

⁷University Hospital of Copenhagen, Copenhagen, Denmark

Abstract

We present a mechanistic interpretability framework for EEG foundation models, combining Sparse Autoencoders (SAEs), spectral decoding, and concept-based attribution to make learned representations legible at the clinical level. Applied to SleepFM finetuned on 434K tokens from 1,714 subjects, we find that layer-wise TopK SAEs decompose encoder activations into features with interpretable spectral signatures; predominantly δ/θ slow waves in the pathology-selective layer. We introduce *concept steering*: substituting concept-enriched SAE feature activations with a target-group centroid and decoding through the spectral decoder, which provides mechanistic, visualisable evidence of what each feature encodes. Critically, we show that the standard monosemanticity criterion (one feature encodes exactly one concept) is *systematically violated* in clinical EEG, and that this violation is not a model failure. We redefine monosemanticity with a three-way taxonomy: separable, entangled, and spurious, and demonstrate via an SAE expansion sweep that entangled polysemanticity (specifically, the age $\times$ pathology co-activation) is biologically irreducible: it persists at all expansion ratios while spurious polysemanticity resolves with capacity. The entanglement is a discovery: the δ/θ subspace underlying infant maturation and pathological slowing is a single neural primitive in the encoder, faithfully exposed by the SAE.

1 Introduction

EEG foundation models pre-trained on large corpora achieve strong performance across a range of tasks: sleep staging [1], pathology detection [2], and brain-computer interface (BCI) decoding [3], [4] — yet the internal computations that produce these predictions remain opaque. Unlike image models where attention maps provide coarse visual explanations, EEG models operate on high-dimensional multichannel time-series whose clinically relevant features — spindles, K-complexes, δ waves, epileptiform discharges — are transient and defined by decades of human expert knowledge [5]. Understanding *what* an EEG foundation model has learned, and *where* it has learned it, is both a scientific question and a clinical trust requirement.

Mechanistic interpretability [6], [7] addresses this by reverse-engineering the functional roles of individual neurons and directions inside a network. The central obstacle is *superposition*: a transformer layer with d dimensions can represent up to $n \gg d$ features by co-activating them sparsely, making individual neurons uninterpretable. Sparse Autoencoders (SAEs) [8], [9], [10] have emerged as the dominant tool for recovering these features in language models, but have not yet been applied to biosignal foundation models.

Clinical EEG introduces a challenge absent from language model interpretability: the concepts encoded by a model are not orthogonal. Concepts like age, pathology, and medication are biologically intertwined — an infant’s δ waves and a patient’s pathological slowing are spectrally identical [11]. A naive application of language-model monosemanticity metrics would label every feature that responds to both age and pathology as “polysemantic” and therefore uninterpretable. We show this interpretation is incorrect.

Contributions. We make the following contributions:

1. **Interpretability pipeline.** A three-stage framework — a spectral decoder for embedding-to-spectrum mapping, layer-wise TopK SAEs for feature extraction, and TCAV [12] for concept attribution — applied to SleepFM [1] finetuned on clinical EEG.
2. **Concept steering.** A novel method that substitutes concept-enriched SAE feature activations with a target-group centroid, then decodes through the spectral decoder, providing mechanistic spectral evidence of what each feature encodes. This extends TCAV from a scalar score to a full spectral signature.
3. **Three-way monosemanticity taxonomy.** We redefine monosemanticity for entangled clinical concepts, distinguishing *separable-monosemantic* (one concept, clean steering), *entangled-monosemantic* (multiple labels sharing one neural primitive, biologically irreducible), and *spurious-polysemantic* (capacity-limited superposition, resolves with more SAE features).
4. **Biological entanglement as a scientific finding.** Via an SAE capacity sweep ($128 \rightarrow 1024$ features), we show that age–pathology co-activation persists at all expansion ratios while spurious polysemanticity resolves, identifying the δ/θ subspace as a shared neural primitive. Concept steering confirms this: child pathological EEG cannot be fully de-noised of its developmental δ by any steering of the pathology concept alone.

2 Background

2.1 EEG Foundation Models

SleepFM [1] is a SetTransformer pre-trained on multi-modal polysomnography (27-channel EEG, ECG, respiratory signals) via a contrastive InfoNCE objective. We study the finetuned variant ($d = 128$, 3 transformer layers), which achieves state-of-the-art pathology detection on our clinical dataset **[TODO: Add model performance metrics]**. *REVE* [2] is a 22-layer cross-lingual EEG transformer ($d = 512$) pre-trained on a large multilingual EEG corpus, providing a higher-capacity comparison point.

2.2 Sparse Autoencoders and Superposition

A TopK SAE [13] trained on activations $\mathbf{a} \in \mathbb{R}^d$ learns:

$$\mathbf{z} = \text{TopK}\left(\mathbf{W}_{\text{enc}}\left(\frac{\mathbf{a}-\boldsymbol{\mu}}{\sigma}\right), k\right), \quad \hat{\mathbf{a}} = \mathbf{W}_{\text{dec}}\mathbf{z} + \mathbf{b}_{\text{dec}}, \quad (1)$$

where $\mathbf{W}_{\text{dec}} \in \mathbb{R}^{d \times n}$ has unit-norm columns (decoder directions $\mathbf{w}_i \in \mathbb{R}^d$). The conventional SAE formulation uses an L_1 sparsity penalty [7], which encourages but does not guarantee sparse activations and requires tuning the penalty coefficient λ . We instead use the TopK hard constraint [9], [13]: pre-activations are computed for all n features, then all but the k largest are set to zero. Exactly k features fire per token by construction, making sparsity directly controllable without additional hyperparameters. The model is trained to minimise MSE reconstruction loss under this constraint. Under the *superposition hypothesis* [6], a layer with d neurons can represent $n \gg d$

near-orthogonal features by co-activating them sparsely; SAEs recover these features as sparse directions.

2.3 TCAV and Concept Attribution

Concept Activation Vectors [12] define a concept C via a linear classifier separating concept examples X_C from negatives $X_{\neg C}$ in representation space, yielding a unit CAV $v_C \in \mathbb{R}^d$. We train a linear probe w on SAE z -features to predict concept membership. The probe assigns each example a scalar sensitivity score $S_C(x)$: how strongly the active SAE features align with v_C , and the TCAV score is simply the fraction of concept examples with positive sensitivity:

$$\text{TCAV}_C = \frac{|\{x \in X_C : S_C(x) > 0\}|}{|X_C|}. \quad (2)$$

A score above 0.5 indicates the concept is encoded; significance is assessed against $N_{\text{rand}} = 50$ random-label null CAVs. TCAV has previously been applied to EEG using the BENDR transformer [14], demonstrating that both externally-labelled and anatomically-grounded concepts yield meaningful attributions in raw activation space. Our work extends this to SAE feature space, replacing neuron-level attribution with sparse decoder directions.

3 Methodology

3.1 Dataset

We use a clinical 27-lead EEG dataset of 1,714 subjects (434K 1-second tokens) from routine EEG recordings. Recordings are all collected with BrainCapture’s BC-1 system, and are annotated with age, sex, clinical indication, medication class, and overall classification (normal / abnormal). Pathology subtypes include epileptiform discharges and other abnormalities such as slow wave activity. Data are split by subject using stratified GroupShuffleSplit (80/20 train/validation) to prevent leakage. EEG is preprocessed [15] at 128 Hz for SleepFM; REVE uses 200 Hz.

3.2 Stage I: Spectral Decoder

The *spectral decoder* is a shallow MLP that maps a frozen encoder token embedding $t \in \mathbb{R}^d$ to amplitude and phase for each frequency bin:

$$\hat{A}_f, (\cos \hat{\phi}_f, \sin \hat{\phi}_f) = \text{SD}(t), \quad f = 1, \dots, F. \quad (3)$$

Phase is parameterised as a unit-vector pair to eliminate the 2π discontinuity. SleepFM tokens are channel-averaged ($F = 64$ bins, 0–64 Hz); REVE tokens are per-channel-per-second ($F = 100$ bins, 0.5–100 Hz). Only amplitude is reliably recoverable ($R^2 \approx 0.80$ for SleepFM; phase $R^2 \approx 0$).

The spectral decoder serves two roles: (i) it provides a physiological vocabulary for interpreting SAE decoder directions — any direction $w_i \in \mathbb{R}^d$ can be decoded to a spectral signature via $\text{SD}(w_i)$ — and (ii) it is the final stage of concept steering (Section 3.5).

3.3 Stage II: Layer-wise SAE Training

For each encoder layer ℓ we collect all available training-split token activations, normalise per-dimension, and train a TopK SAE with $n = E \cdot d$ features, where d is the encoder embedding dimension and E is the *expansion ratio* controlling dictionary size relative to the encoder. The model is trained with sparsity k . The reconstruction objective is MSE on normalised activations; dead neurons (never firing in 500 steps) are periodically resampled following [9]. Hyperparameters are selected via a grid search over $E \in \{1, 2, 4, 8\}$ and $k \in \{4, 8, 16, 32\}$ on held-out validation tokens (see Appendix). Final setting: $E = 1$, $k = 8$ (SleepFM finetuned, layer 2, 128 features). **[TODO: Confirm final E — candidate is E=4 (resolves most spurious polysemanticity without excess inactive features; see §4 expansion sweep). Update all downstream numbers if changed.]**

3.4 Stage III: Concept Attribution via TCAV

For each clinical concept C (age group, sex, indication, medication class, abnormality) we train $K = 10$ -fold L_2 -regularised logistic regression CAVs on layer- ℓ activations, yielding v_C and fold-level accuracy. Feature enrichment uses a z-test on per-feature concept/non-concept firing rates with Benjamini–Hochberg correction ($q < 0.05$) at the subject level. For the abnormality concept, classification labels are noisy at the token level; we therefore rank features by the difference in firing rate between abnormal and normal tokens (Δ -firing-rate), which is more robust than binary enrichment at the token level.

3.5 Stage IV: Concept Steering

Concept steering provides interventional evidence of what a set of concept-enriched features encodes. Given source tokens X_{source} (e.g., child abnormal EEG), target tokens X_{target} (e.g., adult normal EEG), and a ranked list of concept-enriched feature indices $\mathcal{F}_C$:

1. Compute per-token SAE features: $z = \text{SAE}(\mathbf{a})$ for every token in both X_{source} and X_{target} .
2. Compute the target centroid – the *mean* SAE feature vector across all target tokens:

$$\mathbf{c}_T = \frac{1}{|X_{\text{target}}|} \sum_{\mathbf{x} \in X_{\text{target}}} \mathbf{z}(\mathbf{x}) \in \mathbb{R}^n. \quad (4)$$

This is the “typical” activation pattern of group X_{target} in SAE space.

3. For each source token, selectively substitute the n most concept-enriched feature activations with the corresponding centroid values ($n \in \{0, 20, 50\}$): $z_i^* = \begin{cases} c_{T,i} & \text{if } i \in \mathcal{F}_C^{\text{top-}n} \\ z_i & \text{otherwise.} \end{cases}$
4. Decode via SAE decoder: $\hat{\mathbf{a}}^* = \mathbf{W}_{\text{dec}} \mathbf{z}^* + \mathbf{b}_{\text{dec}}$.
5. Decode via spectral decoder: $\hat{A}^*(f) = \text{SD}(\hat{\mathbf{a}}^*)$, recover amplitude spectrum.

The substitution asks: “if the top- n concept features looked like they do in a typical target-group token, what spectrum would result?”. The resulting curve $\hat{A}^*(f)$ is compared against the target reference band (mean ± 1 SD across X_{target}). Full convergence at small n implies those features

mechanistically and completely encode the concept; a persistent residual after $n = 50$ implies entanglement, meaning that the concept cannot be isolated along a single feature dimension.

3.6 Monosemanticity Taxonomy

Standard monosemanticity [8] defines a feature as monosemantic if it responds to exactly one interpretable concept. For SAE feature $i \in \{1, \dots, n\}$, the canonical metric is *field breadth*:

$$B(i) = |\{m \in \mathcal{M} : \exists c \in \mathcal{C}_m \text{ s.t. } p_{i,m,c}^{\text{BH}} < 0.05\}|, \quad (5)$$

where $\mathcal{M}$ is the set of metadata fields (age group, sex, indication, medication, pathology), $\mathcal{C}_m$ is the set of discrete categories within field m , and $p_{i,m,c}^{\text{BH}}$ is the BH-corrected enrichment p -value for feature i firing preferentially on tokens from category c of field m . Monosemantic: $B(i) = 1$; polysemantic: $B(i) \geq 2$.

This definition assumes clinical concepts are orthogonal. We relax this with a three-way taxonomy. Let $\mathcal{E} = \{\text{age group, classification, medication, indication}\}$ be the *entanglement cluster*, that is, fields that are biologically co-determined (age shapes pathology expression; diagnosis drives medication). Then:

- **Inactive:** $B(i) = 0$ (feature never fires significantly).
- **Separable-monosemantic:** $B(i) = 1$.
- **Entangled-monosemantic:** $B(i) \geq 2$ and all fields contributing to $B(i)$ lie within $\mathcal{E}$.
- **Spurious-polysemantic:** $B(i) \geq 2$ and at least one field contributing to $B(i)$ lies outside $\mathcal{E}$.

Entangled-monosemantic features are not a failure of the SAE — they are a discovery that the encoder represents multiple clinical labels via a single shared neural primitive.

[TODO: Figure out exactly how to define $\mathcal{E}$, talk with Sandor]

4 Results

4.1 The Interpretability Framework

Spectral decoder reconstruction. The spectral decoder achieves $R^2 \approx 0.80$ for amplitude reconstruction on held-out SleepFM tokens, with performance degrading above 40 Hz where token embeddings carry little spectral content. Phase is not recoverable from the embedding ($R^2 \approx 0$), consistent with the encoder’s contrastive objective discarding phase information.

SAE feature spectra. After decoding each SAE decoder direction through the spectral decoder, SAE features resolve into narrow spectral signatures. The dominant features at layer 2 peak in δ (0.5–4 Hz) and θ (4–8 Hz), consistent with the layer’s role in encoding clinical abnormality (pathological slowing) and age (infant δ dominance).

SAE reconstruction preserves task performance. As a faithfulness check, we replace the encoder’s layer-2 activations with their SAE reconstructions and re-evaluate the downstream normal/abnormal classifier. Classification performance does not drop ([TODO: insert AUROC before/after]), confirming that the SAE captures the task-relevant structure of the representation without discarding diagnostically meaningful variance.

TCAV concept emergence. Model-level TCAV scores (Figure 1) reveal a clear hierarchy: spectral primitives are encoded before clinical concepts. δ and θ are significant at layer 0 ($\text{TCAV}_\delta = 1.00$, $p < 0.001$), while the abnormality concept starts near chance ($\text{TCAV}_{\text{abn.}} = 0.53$, $p = 0.44$) and reaches 0.97 only at layer 2 ($p < 0.001$). Note that SleepFM is finetuned on abnormality classification, making its strong representation in the final layer expected; spectral concepts such as δ and θ are not finetuning targets, so their significance from layer 0 onward reflects structure learned during self-supervised pre-training. [TODO: Update TCAV text and caption once metadata concepts (child, female, ASM, seizure) are added to the cache — rewrite “all five measurable concepts” claim accordingly]

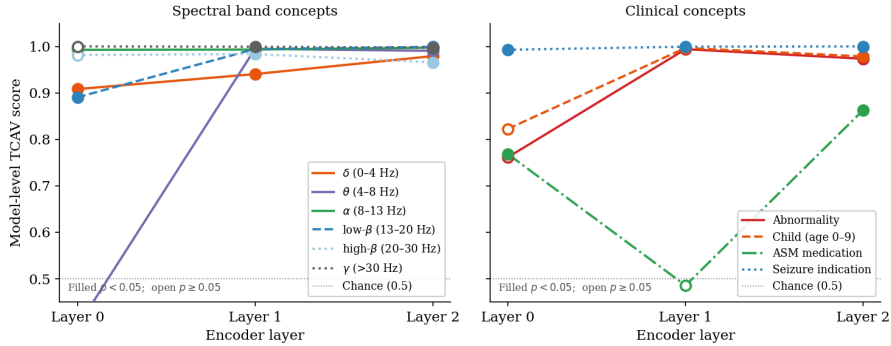


Figure 1: **TCAV concept emergence across SleepFM layers.** Model-level TCAV scores for spectral bands and clinical concepts. Filled markers: $p < 0.05$ vs. random-label null (50 permutations). Open markers: $p \geq 0.05$. Spectral primitives (δ , θ) are already significant at layer 0; clinical abnormality and metadata concepts emerge later, confirming a coarse-to-fine encoding hierarchy. [TODO: Regenerate figure once TCAV re-run with metadata concepts completes]

4.2 Redefining Monosemanticity for Clinical EEG

The three-way taxonomy. Applying the taxonomy of Section 3.6 to SleepFM finetuned layer 2 (128 features, $k = 8$):

The cross-field co-occurrence heatmap (Figure 2, left) shows that age-classification co-activation (diagonal block of $\mathcal{E}$) accounts for the majority of entangled features, with a co-occurrence prevalence of 0.78.

The expansion sweep distinguishes biological from capacity constraints. Figure 2 (right) shows taxonomy fractions as a function of SAE expansion ratio. Spurious polysemanticity drops monotonically: 25% → 22% → 15% → 7% at 8× expansion. This is the capacity-limited component — more dictionary entries allow the SAE to assign separate features to unrelated concepts that were previously superposed.

Entangled polysemanticity is stable or increases: 45% → 50% → 52% → 45%. It does *not* resolve with expansion. This confirms the biological constraint: age and pathology share a neural primitive (δ/θ slow waves) that no amount of SAE capacity can separate into independent features, because the encoder itself does not represent them independently.

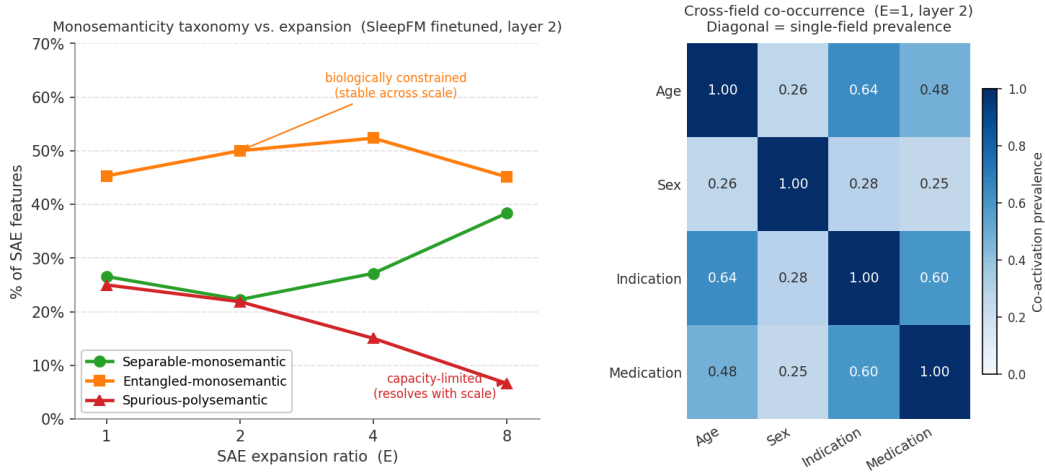


Figure 2: **Left:** Feature taxonomy fractions vs SAE expansion ratio (SleepFM finetuned, layer 2). Spurious polysemanticity (red) decreases monotonically from 25% to 7% as capacity grows — a hallmark of superposition under dictionary shortage. Entangled polysemanticity (orange) remains stable at 45–52%, confirming its biological rather than capacity origin. **Right:** Cross-field co-occurrence heatmap showing why entanglement is stable: age, indication, and medication share a neural primitive (δ/θ slow waves) that no SAE dictionary size can separate, because the encoder itself does not represent them independently. Sex co-occurrence is substantially lower (0.25–0.26), consistent with its separable character.

Key finding. Expansion=4 is the recommended operating point: spurious polysemanticity is largely resolved (15%) while the irreducible entanglement remains visible and characterisable. Expansion=8 adds inactive features (10%) without further reducing biological entanglement.

4.3 Concept Steering as Evidence for Entanglement

Concept steering (Section 3.5) provides direct spectral evidence for the taxonomy. The three-way taxonomy predicts qualitatively different steering behaviour: separable features steer cleanly with no residual; entangled features leave a residual that cannot be removed by steering a single concept; spurious features produce noisy or null responses.

Age steering confirms separability in healthy brains. Substituting the top-50 age-enriched features with the 50+ yr centroid moves child normal EEG fully into the adult normal reference band (Figure 3, top), including the δ band. The steered spectrum falls within the adult mean ± 1 SD at all frequencies. This confirms age features are separable-monosemantic in healthy EEG: they encode the spectral maturation trajectory without entanglement from pathology.

Child/adult split reveals asymmetric entanglement. Figure 3 (bottom) shows the critical dissociation. Adult abnormal EEG steers cleanly to adult normal (right panel): pathological δ is removed and a posterior α peak emerges. Child abnormal EEG retains residual δ above the child-normal reference after $n = 50$ substitutions (left panel): the developmental δ is entangled with

pathological δ in the feature representation, and cannot be removed via pathology features without also removing the developmental signal.

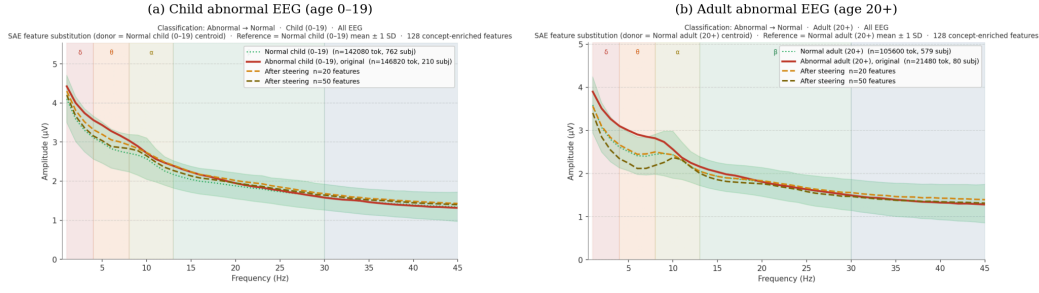


Figure 3: Concept steering reveals asymmetric age-pathology entanglement. Progressive substitution of age-enriched SAE features ($n = 0 \dots 50$) in abnormal EEG. **(a) Child (0–19):** Residual δ above the child-normal reference persists at $n = 50$ — developmental δ is entangled with pathological δ and cannot be removed via pathology features alone. **(b) Adult (20+):** Clean removal — pathological δ is separable from age effects and the posterior α peak re-emerges after steering.

Probe curve confirms shared neural primitive. The maturation-pathology Pareto curve (Figure 4) shows that age AUROC (measuring age decodability) and clinical AUROC (pathology decodability) collapse jointly when age-enriched features are progressively clamped. The first ≈ 50 features are orthogonal maturation markers (age AUROC drops, clinical AUROC is flat); beyond this, the same features are load-bearing for both age and pathology decoding. This quantitatively confirms that a shared δ/θ neural primitive underlies both developmental maturation and pathological slowing.

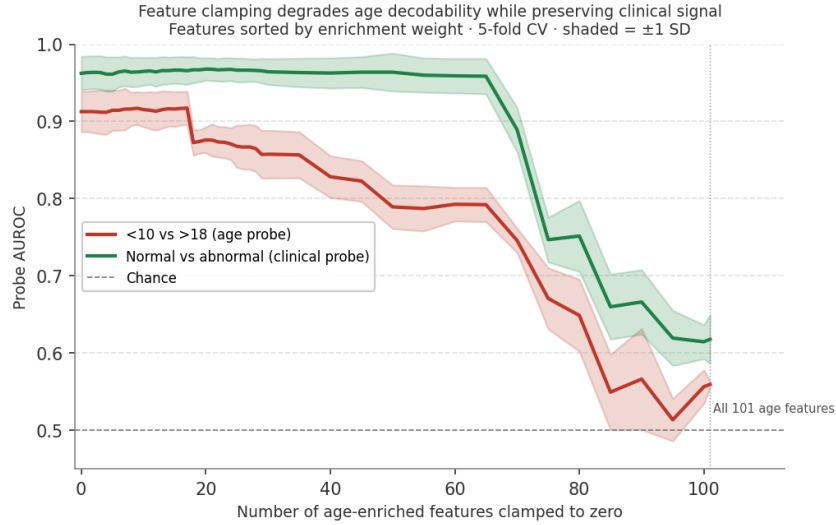


Figure 4: Maturation-Pathology Pareto Front. Age AUROC (red) and clinical AUROC (green) as features are progressively clamped. The joint collapse beyond feature ≈ 50 identifies the entangled δ/θ subspace shared between infant maturation and pathological slowing.

Epileptiform-diffuse slowing null result. Zero SAE features significantly discriminate epileptiform from non-epileptiform pathology (BH $q < 0.05$). Both subtypes produce the same δ/θ spectral representation at layer 2, indicating that epileptiform-specific morphology (spike morphology, ictal dynamics) is not encoded as a separable dimension in SleepFM layer 2. This is an entanglement at the pathology subtype level, consistent with the model’s spectral rather than morphological feature representation. **[TODO: This may be due to me collapsing the two forms of abnormalities when finetuning the model. I would like to rerun this with abnormality types labelled explicitly.]**

Steering quality is stable across expansion ratios. Repeating age steering at expansion $1\times$, $2\times$, $4\times$, and $8\times$ produces visually indistinguishable spectral curves (Figure 5). The top- n features at expansion $1\times$ already capture the age concept as effectively as 1024-feature SAEs. This is consistent with the age feature being separable-monosemantic — it occupies a low-dimensional subspace fully resolved by 128 features.

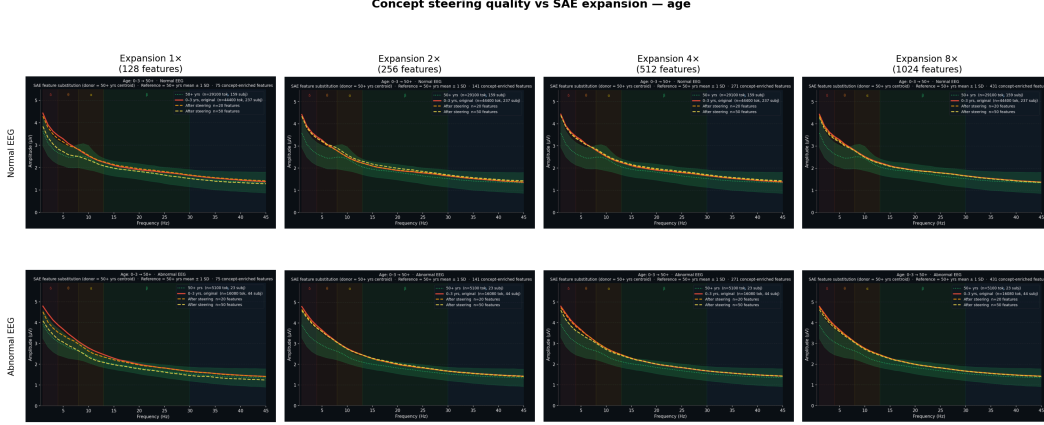


Figure 5: **Age steering quality across SAE expansion ratios.** Top row: normal EEG; bottom row: abnormal EEG. Columns: $1\times/2\times/4\times/8\times$ expansion (128–1024 features). The steered spectral curves are indistinguishable across expansions — the age concept saturates at 128 features.

5 Discussion

Entanglement as biological discovery. The central finding is that the age–pathology polysemanticity in SleepFM layer 2 is not a failure of the SAE or of the model — it is an accurate reflection of the encoder’s representation. The model has learned that developmental δ and pathological δ are, at the spectral level, the same neural primitive. The SAE faithfully exposes this shared representation. The correct conclusion is not “the model is uninterpretable” but “the model encodes a biologically real fact: infant EEG and pathological adult EEG are spectrally similar, and the encoder cannot distinguish them without non-spectral information.” This has direct clinical implications: spectral-only models will systematically confuse young-child abnormality with age-normal variation.

Concept steering as a new interpretability primitive. Existing TCAV methods [12], [14] produce scalar attribution scores. Concept steering extends this to a full spectral signature, enabling qualitative assessment of what a feature encodes and whether a concept is separable. The child/adult split in steering results (Figure 3) could not have been identified by AUROC scores alone — the spatial pattern of residual δ uniquely identifies the entanglement mechanism.

Limitations. Phase information is not recoverable from SleepFM embeddings, restricting our spectral analysis to amplitude. The entanglement cluster $\mathcal{E}$ was defined a priori based on biological reasoning; future work should test whether other field combinations exhibit similar non-orthogonality. The epileptiform null result is specific to SleepFM layer 2 and may not generalise to models trained with morphological objectives.

[MAYBE: Add REVE cross-model comparison here once REVE steering is run]

6 Conclusion

We introduced a mechanistic interpretability framework for EEG foundation models combining a spectral decoder, sparse feature extraction (SAE), concept attribution (TCAV), and a novel concept steering method. Applied to SleepFM finetuned on 434K clinical EEG tokens, we found that the standard monosemanticity criterion systematically misclassifies biologically entangled features as polysemantic. Our three-way taxonomy distinguishes separable (capacity-resolvable) from

entangled (biologically irreducible) polysemanticity, confirmed by an SAE expansion sweep and by direct spectral steering experiments. The age–pathology co-activation in δ/θ SAE features is a scientific finding: the encoder represents developmental maturation and pathological slowing as a shared neural primitive, with direct implications for clinical model auditing and the interpretation of spectral AI features in routine EEG practice.

References

- [1] R. Thapa and others, “SleepFM: Multi-modal Representation Learning for Sleep Across Brain Activity, ECG and Respiratory Signals,” in *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.
- [2] Y. El Ouahidi *et al.*, “REVE: A Foundation Model for EEG: Adapting to Any Setup with Large-Scale Pretraining on 25,000 Subjects,” *Advances in Neural Information Processing Systems*, 2025, [Online]. Available: <https://brain-bzh.github.io/reve/>
- [3] D. Kostas, S. Aroca-Ouellette, and F. Rudzicz, “BENDR: Using Transformers and a Contrastive Self-Supervised Learning Task to Learn from Massive Amounts of EEG Data,” *Frontiers in Human Neuroscience*, vol. 15, 2021.
- [4] W.-B. Jiang, L.-M. Zhao, and B.-L. Lu, “Large Brain Model for Learning Generic Representations with Tremendous EEG Data in BCI,” in *International Conference on Learning Representations (ICLR)*, 2024.
- [5] S. Beniczky and others, “Standardized Computer-based Organized Reporting of EEG: SCORE,” *Epilepsia*, 2020.
- [6] N. Elhage and others, “Toy Models of Superposition,” *Transformer Circuits Thread*, 2022.
- [7] H. Cunningham and others, “Sparse Autoencoders Find Highly Interpretable Features in Language Models,” *arXiv preprint arXiv:2309.08600*, 2023.
- [8] T. Bricken and others, “Towards Monosemanticity: Decomposing Language Models with Dictionary Learning,” *Transformer Circuits Thread*, 2023.
- [9] A. Templeton and others, “Scaling and Evaluating Sparse Autoencoders,” *Anthropic Technical Report*, 2024.
- [10] T. Lieberum and others, “Gemma Scope: Open Sparse Autoencoders Everywhere All At Once on Gemma 2,” *arXiv preprint arXiv:2408.05147*, 2024.
- [11] E. Niedermeyer and F. H. Lopes da Silva, Eds., *Electroencephalography: Basic Principles, Clinical Applications, and Related Fields*, 5th ed. Lippincott Williams & Wilkins, 2005.
- [12] B. Kim *et al.*, “Interpretability Beyond Classification Accuracy: Quantitative Testing with Concept Activation Vectors (TCAV),” in *Proceedings of the 35th International Conference on Machine Learning (ICML)*, 2018.
- [13] A. Makhzani and B. Frey, “k-Sparse Autoencoders,” in *International Conference on Learning Representations (ICLR)*, 2014.
- [14] A. G. Madsen, W. T. Lehn-Schiøler, Á. Jónsdóttir, B. Arnardóttir, and L. K. Hansen, “Concept-Based Explainability for an EEG Transformer Model,” in *2023 IEEE 33rd International Workshop on Machine Learning for Signal Processing (MLSP)*, IEEE, Sept. 2023, pp. 1–6. doi: 10.1109/mlsp55844.2023.10285992.
- [15] A. Gjørbye, L. Skerath, W. Lehn-Schiøler, N. Langer, and L. K. Hansen, “SPEED: Scalable Preprocessing of EEG Data for Self-Supervised Learning,” in *Proceedings of the 2024 IEEE International Workshop on Machine Learning for Signal Processing*, 2024.